

# KLS Gogte Institute of Technology, Belagavi

## IA Test – I

Program: BE

Department: Information Science & Engineering

Course Title: Data Structures and Applications

Duration: 1 Hr.

Semester: III

Academic Year: 2024-25 (ODD SEM)

Max. Marks: 30 marks

Code: 22IS34

Date: 26/09/24

|               |                                                      |
|---------------|------------------------------------------------------|
| Instructions: | Write comments in the programs where ever necessary. |
|---------------|------------------------------------------------------|

| Q.No                            | PART A (Compulsory question)                                                                                                                                                                                                                                                                                                                                      | [L] | [CO] | [PO] | [M] |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|------|------|-----|
| 1a.                             | Define Outcome Based Education (OBE).                                                                                                                                                                                                                                                                                                                             | 2   |      | 1    | 1   |
| 1b.                             | Define CO and PO.                                                                                                                                                                                                                                                                                                                                                 |     |      |      | 1   |
| 1c.                             | Define data structure. Give examples for two data structures with their applications.                                                                                                                                                                                                                                                                             |     |      |      | 4   |
| 1d.                             | Explain the following file handling functions of C: fopen, feof, fscanf and fprintf along with their syntax.                                                                                                                                                                                                                                                      |     |      |      | 4   |
| PART B (Answer any 1 questions) |                                                                                                                                                                                                                                                                                                                                                                   |     |      |      |     |
| 2.                              | Develop a C program to read and display the information namely, Name, USN , Branch, Sem and CGPA) of 50 students using structure and display the topper of the class.                                                                                                                                                                                             | 3   | 2    | 3    | 10  |
| 3.                              | Develop a C program to implement stack operations.                                                                                                                                                                                                                                                                                                                | 3   | 2    | 3    | 10  |
| PART C (Answer any 1 question)  |                                                                                                                                                                                                                                                                                                                                                                   |     |      |      |     |
| 4.                              | Covert the following Infix expression in to the postfix expression using tabulation method and also display the content of the stack.<br>i) $(A+B*C-D/(E+F)/G)$<br>ii) $((5+4)*6)/2+(7-3)$                                                                                                                                                                        | 3   | 3    | 1    | 10  |
| 5.                              | During the promotion of students from second semester to third semester student data is captured and stored in a input file, which contains information of 'N' students (Name, USN, Branch, Semester and CGPA). Develop a C program to transfer the contents of input file to output file such that the student records are stored in an ascending order of CGPA. | 4   | 2    | 3    | 10  |

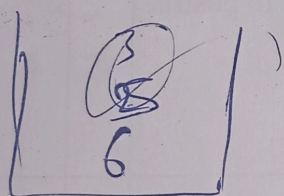
|               |   |                                                |
|---------------|---|------------------------------------------------|
| Instructions: | 1 | Missing data, if any, may be suitable assumed. |
|---------------|---|------------------------------------------------|

| Q.No                            | PART A (Compulsory question)                                                                                                                                                                                                                                                                                                                                                                                                                     | [L] | [CO] | [PO]  | [M]         |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|------|-------|-------------|
| 1.                              | <p>a. Name the Program Outcomes (POs) ,which are related to the fundamental operations of a stack, such as push, pop, and display?</p> <p>b. What is Structure in C? Explain the different ways of declaring structures with examples.</p> <p>c. Write C program to find sum of elements of array using pointer.</p>                                                                                                                             | 2   | 1,2  | 1,2,3 | 2<br>5<br>3 |
| PART B (Answer any 1 questions) |                                                                                                                                                                                                                                                                                                                                                                                                                                                  |     |      |       |             |
| 2.                              | <p>Identify the data structure required to convert the given infix expression to postfix expression and apply the same to convert the following expression.</p> <p><math>(A + B) * (C - (D / E \wedge F) * G) + H</math></p>                                                                                                                                                                                                                     | 3   | 3    | 2,3   | 10          |
| 3.                              | <p>Develop a program in C that defines a structure for Employee containing employee ID, name, and salary. The program should allow the user to input details of N employees and then sort them by salary in ascending order.</p>                                                                                                                                                                                                                 | 3   | 1    | 2     | 10          |
| PART C (Answer any,1 question)  |                                                                                                                                                                                                                                                                                                                                                                                                                                                  |     |      |       |             |
| 4.                              | <p>You are working in a logistics company that manages shipments. Each shipment is processed in a First In, First Out (FIFO) manner. Your task is to examine how the process works in different operations and track the status after each action. The maximum capacity of the queue is 3:</p> <p>i. display() →</p> <p>ii. enqueue(2)</p> <p>iii. enqueue(4)</p> <p>iv. dequeue()</p> <p>v. rear()</p> <p>vi. enqueue(8)</p> <p>vii. peek()</p> | 4   | 3    | 2,3   | 10          |

|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                    |   |   |     |    |
|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|---|---|-----|----|
|    | viii. enqueue(7)<br>ix. dequeue()<br>x. enqueue(11)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <div style="border: 1px solid black; padding: 2px; display: inline-block;"> 2502-1<br/> 317 </div> |   |   |     |    |
| 5. | <p>You are working as a software developer for a company that builds a game AI. The AI needs to evaluate mathematical expressions provided in postfix notation to make real-time decisions during gameplay. Since postfix expressions don't need parentheses to handle operator precedence, it's ideal for the fast-paced environment in which the AI operates.</p> <p>You've been given the following postfix expression that represents a calculation needed for the AI to decide the next move. Your task is to analyze this expression step-by-step, using a suitable data structure, and arrive at the result so that the next move can be decided.</p> <p style="text-align: center;"><math>6\ 2\ 3\ +\ -\ 3\ 8\ 2\ /\ +\ *</math></p> |                                                                                                    | 4 | 3 | 2,3 | 10 |

$$rear = (rear + 1) \% 5$$

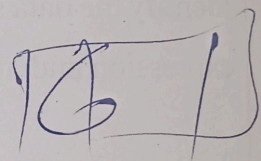
3 | 4 | 5 | 1 | 2



$$2 + 3 = 5$$

$$6 - 5 = 1$$

$$rear = (rear + 1) \% 5$$



Sum

6  
2  
3  
+  
-  
3  
8  
2  
/

Not Exp

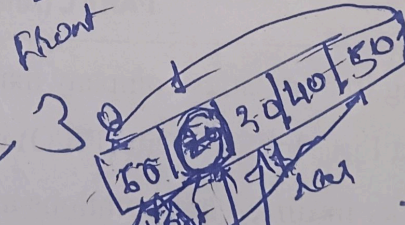
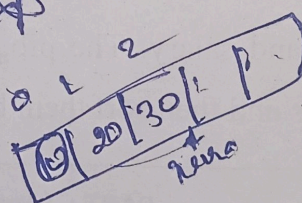
6

6, 2

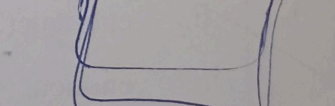
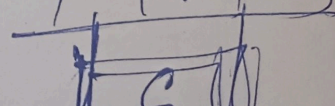
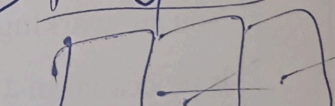
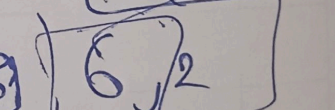
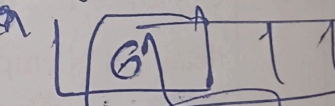
6, 2, 3

6, 2, 3, 5

6, 5, 3



Stack



$$rear = 0$$

$$front = 0$$

$$5 - 5 = 0$$

**IA Test – II**

Program: B.E

Department: CSE

Course Title: Data Structures and Applications

Duration: 1 Hr.

Academic Year: 2024-25 (ODD SEM)

Semester: 3

Max. Marks: 30 marks

Code: 22CS34

Date: 15/11/2024

|               |   |                                                                            |
|---------------|---|----------------------------------------------------------------------------|
| Instructions: | 1 | Missing data, if any, may be suitable assumed.                             |
|               | 2 | Part A is compulsory.<br>Answer any 1 question from Part B and Part C each |

| Q.No                                   | PART A (Compulsory question)                                                                                                                                                                                                                                                                                                                                                                                                                                                            | [L] | [CO] | [PO] | [M] |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|------|------|-----|
| 1a                                     | A person wants to develop a program to illustrate the message passing system using a suitable data structure. His task relates to which course outcome                                                                                                                                                                                                                                                                                                                                  | 1   | 3    | 2,3  | 2   |
| 1b.                                    | <p>a. Define binary search tree. Answer the following with respect to the tree given.</p> <p>a. Root of the tree</p> <p>b. Height of the tree</p> <p>c. Level of the tree</p> <p>d. Balance factor</p> <p>e. Sibling of B</p> <p>f. Degree of D</p> <p>g. Ancestor of G</p> <p>h. Descendants of B</p> <p style="text-align: center;"><b>Tree</b></p>                                                                                                                                   | 2   | 2    | 2,3  | 8   |
| <b>PART B (Answer any 1 questions)</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |     |      |      |     |
| 2                                      | Develop a program to implement the basic operations of circular queue.                                                                                                                                                                                                                                                                                                                                                                                                                  | 3   | 3    | 2,3  | 10  |
| 3                                      | Develop a C program to traverse an ordered list implemented using a linked list and delete all the nodes whose keys are odd numbers.                                                                                                                                                                                                                                                                                                                                                    | 3   | 3    | 2,3  | 10  |
| <b>PART C (Answer any 1 question)</b>  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |     |      |      |     |
| 4.                                     | <p>A library management system needs to organize book information using a Binary Search Tree (BST) based on the unique book IDs. Your task is to assist the librarian in arranging the books by inserting the following book IDs into a BST. After the tree is constructed, you will need to develop functions to perform different tree traversals (In-order, Pre-order, and Post-order) and display the books in various orders.</p> <p>58,96,45,76,59,42,31,68,42,25,61,18,20,78</p> | 4   | 4    | 2,3  | 10  |

|     |                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |   |     |   |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|-----|---|
| 5   | <p>Consider the tree given below as input for questions 5a and 5b.</p> <pre> graph TD     42((42)) --&gt; 28((28))     42 --&gt; 62((62))     28 --&gt; 21((21))     28 --&gt; 31((31))     21 --&gt; 17((17))     21 --&gt; 23((23))     31 --&gt; 30((30))     31 --&gt; 35((35))     62 --&gt; 59((59))     62 --&gt; 71((71))     59 --&gt; 57((57))     59 --&gt; 60((60))     71 --&gt; 67((67))     71 --&gt; 82((82)) </pre> |   |   |     |   |
| 5a. | <p>Analyze the following code snippet and write the output for the given input tree.</p> <pre> struct Node* func(struct Node* node) {     if (node == NULL) {         printf("The tree is empty.\n");         return NULL;     }      struct Node* current = node;     while (current-&gt;left != NULL) {         current = current-&gt;left;     }     return current; } </pre>                                                     | 4 | 3 | 2,3 | 5 |
| 5b. | <p>Analyze the following code snippet and write the output for the given input tree.</p> <pre> struct Node* func (struct Node* node) {     if (node == NULL) {         printf("The tree is empty.\n");         return NULL;     }      struct Node* current = node;     while (current-&gt;right != NULL) {         current = current-&gt;right;     }      return current; } </pre>                                                 | 4 | 3 | 2,3 | 5 |

Pre order - Root    Left    Right  
 In order - Left    Root    Right  
 Post order - Left    Right    Root